[METRIC]
A-A-59532/2A
March 28, 2006
SUPERSEDING
A-A-59532/2
January 14, 2000

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

FUSE, INCLOSED LINK, SURFACE MOUNT (SM), 63 VOLTS (V) AC/DC, QUICK ACTING WITH FUSEHOLDER, BLOCK

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring the fuse assemblies described herein shall consist of this document and the issue in effect of CID A-A-59532.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59532	/02	-	001
CID number	Applicable CID specification sheet (two digits)		Dash number

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Fuse assemblies supplied to this CID shall be as specified herein (see figure 1).

Electrical specifications.

Ampere rating. The ampere rating shall be as specified in table I.

Voltage rating. The voltage rating shall be 63 V ac/dc.

Time current characteristics. The time current characteristics shall be as indicated in table II.

Environmental specifications. Fuse assemblies supplied to this CID shall be subject to the following tests and there shall be no electrical or mechanical damage to the fuse.

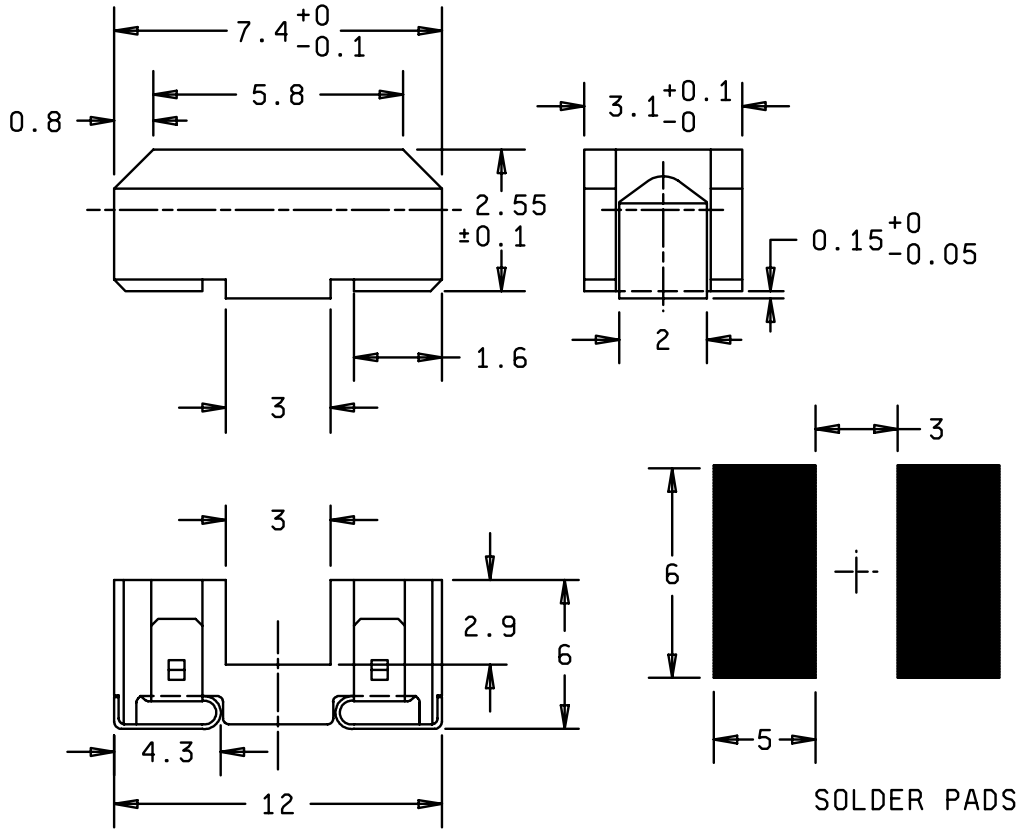
Maximum storage temperature. The maximum storage temperature shall be 40°C (104°F) at 70 percent relative humidity.

Ambient temperature maximum. The ambient temperature shall be -40°C (-40°F) to +85°C (185°F).

Vibration resistance. Fuses shall meet vibration requirements in accordance with IEC 60068-2-6, Test Fc. (10-2000 Hz, cross-over frequency 60 Hz, resp. acceleration 100 m/s^2 (10g)).

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mm	Inches	mm	Inches
0.05	.002	3	.118
0.1	.004	3.1	.122
0.15	.006	4.3	.169
0.8	.031	5	.197
1.6	.063	5.8	.228
2.0	.079	6	.236
2.55	.100	7.4	.291
2.9	.114	12	.472

NOTES:

1. Dimensions are in millimeters.
2. Tolerance is ± 0.13 mm (.005 inch), unless otherwise specified.

FIGURE 1. Configuration and dimensions.

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TABLE I. Electrical characteristics.

AA59532/02-	Ampere rating	Breaking capacity	Voltage drop at rated current. typical mV	Power dissipation at rated current. typical mW	Fusing I^2t at 4 times rated current.	
					Typical A^2s	Maximum A^2s
001	63 m A	50A/ 63V ac/dc p.f./cos w 1	2550	160	0.00011	0.00064
002	100 m A		1770	180	0.0067	0.0016
003	125 m A		1770	220	0.0011	0.0025
004	160 m A		1700	270	0.0018	0.0041
005	250 m A		990	250	0.0058	0.01
006	350 m A		990	350	0.0076	0.02
007	375 m A		990	370	0.013	0.023
008	400 m A		960	380	0.016	0.026
009	500 m A		350	180	0.01	0.04
010	630 m A		290	180	0.02	0.064
011	750 m A		260	200	0.031	0.09
012	1 A		220	220	0.078	0.16
013	1.25 A		220	280	0.14	0.25
014	1.5 A		200	300	0.24	0.36
015	1.6 A		200	320	0.27	0.41
016	2 A		200	400	0.44	0.64
017	2.5 A		190	480	0.97	1.0
018	3 A		190	570	1.3	1.4
019	3.15 A		190	600	1.2	1.6
020	3.5 A		140	490	1.6	2.0
021	4 A		140	560	2.1	2.6
022	5 A		140	700	2.9	4.0

TABLE II. Opening time characteristics.

Rated current	100 percent of rated current	200 percent of rated current	400 percent of rated current
63 mA - 5A	≥ 4 hours	< 1 second	< 10 milliseconds
6.3A - 8A	≥ 4 hours	< 5 seconds	< 50 milliseconds
10A	≥ 4 hours	< 20 seconds	< 60 milliseconds

Shock resistance. Fuses shall meet the shock requirements in accordance with IEC 60068-2-27, test Ea (981 m/s^2, 6ms).

Solderability (reflow and wave soldering). Fuse assemblies shall be able to withstand, without electrical or mechanical damage to the fuse, a wave or reflow solder of +235°C (455°F) for 2 seconds maximum in accordance with IEC 60068-2-58 test Td.

Resistance to soldering heat. Fuse assemblies shall meet resistance to soldering heat requirements in accordance with IEC 60068-2-58 test Td (10 seconds at 260°C (500 °F)).

Fuse-link temperature rise ≤ 75 K. PC board track widths shall be ≤ 5 mm (0.196 in.) for fuses rated at 5 A or less, ≥ 5 mm (0.196 in.) for fuses rated at 6.3 and 7 A and ≥ 10 mm (.394 in.) for fuses rated at 8 and 10 A.

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Permissible continuous operating current at 23°C (73.5 °F). The permissible continuous operating current at 23°C (73.5 °F) shall be 0.7 times rated current.

Physical specifications.

Materials.

Housing material. The fuses and fuseholder blocks shall be temperature resistance plastic with a UL94 VO flammability rating as a minimum.

Terminal plating. Terminals shall be tin-plated brass. Use of pure tin plating is prohibited as a finial finish and as an undercoat. Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

Additional fuses and fuseholders. The fuses and fuseholder blocks described in this CID may be supplied separately. Additional Fuses may be obtained under A-A-59532/1 and additional fuseholders may be obtained under A-A-59533.

PACKAGING. Packaging shall be as specified in A-A-59532. In addition, these fuse-assemblies may be supplied individually or in a quantity of 1,500 on a 22.4 mm (.881 inches) wide tape reel in accordance with IEC 60286-3.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Source of documents.

Commercial Item Description

A-A-59532/1	-	Fuse, Inclosed Link, Surface Mount (SM), 63 Volts (V) AC/DC, Quick Acting.
A-A-59533	-	Fuseholder Block, 5 Ampere (A), 125 Volt (V) AC/DC, Surface Mount (SM), General Requirements for.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://dodssp.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Publications

Non Government Publications

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60068-2-6	-	Environmental Testing – Part 2: Tests - Test Fc: Vibration (Sinusoidal).
IEC 60068-2-27	-	Environmental Testing – Part 2: Tests - Test Ea and Guidance: Shock.
IEC 60068-2-58	-	Environmental Testing – Part 2-58: Tests - Test Td - Test Methods for Solderability, Resistance to Dissolution of Metallization and to Soldering Heat of Surface Mounting Devices (SMD).
IEC 60286-3	-	Packaging of Components for Automatic Handling - on Continuous Tapes.

(Copies of these documents are available online at http://www.iec.ch/ or from the International Electrotechnical Commission, 3 Rue De Varembe', P.O. Box 131, CH - 1211, Geneve 20, Switzerland.)

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UNDERWRITERS LABORATORIES, INCORPORATED (UL)

UL94 - Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, Standard For.

(Copies of this document is available online at http://www.ul.com/ or from the Underwriters Laboratories, Incorporated, 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
61935	Schurter, Incorporated 447 Aviation Boulevard Santa Rosa, CA 95403-1069 Phone number (707) 636-3000 Facsimile number (707) 636-3033 E-mail: mailto:info@schurterinc.com Uniform Resource Locator (URL): http://www.schurterinc.com

Part number (P/N) supersession data. These CID PINs supersede the following MFR's P/N's as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

CID dash number (see table I)	MFR's CAGE	MFR's P/N 1/	CID dash number (see table I)	MFR's CAGE	MFR's P/N 1/
AA59532/02-			AA59532/02-		
001	61935	3422.0003.11	012	61935	3422.0009.11
002	61935	3422.0004.11	013	61935	3422.0010.11
003	61935	3422.0049.11	014	61935	3422.0047.11
004	61935	3422.0005.11	015	61935	3422.0011.11
005	61935	3422.0006.11	016	61935	3422.0012.11
006	61935	3422.0043.11	017	61935	3422.0013.11
007	61935	3422.0044.11	018	61935	3422.0014.11
008	61935	3422.0007.11	019	61935	3422.0048.11
009	61935	3422.0045.11	020	61935	3422.0015.11
010	61935	3422.0008.11	021	61935	3422.0016.11
011	61935	3422.0046.11	022	61935	3422.0017.11

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown. For actual part marking requirements see the marking paragraph of A-A-59532.

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MILITARY INTERESTS:

Custodians:
Navy - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

DLA-CC

Project 5920-2006-010

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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